وثيقة المراجعة والتلخيص الأكاديمي

ASE · Chapter 5 · Software Design — Aggarwal & Singh, 3rd ed. (Bloom Level 6)

Source: slide deck Ch05_Software_Design.pdf (99 slides) + printed book pp. 202–251. Case study = **Library Management System** (slides 81–93; book pp. 236–247). Chapter data tables: slide 19 (coupling), slide 27 (cohesion), slide 57 (SDD Table 2), slide 72/75 (OOD steps, 6 boxes), slide 75 (object types), slides 77–78 (class relationships), slides 84–93 (case-study artefacts).

Format: English question + full worked artefact in English; one short Arabic justification per item. Every sub-question states its assumptions, produces the artefact (tables / indented trees / ASCII-structured lists), then justifies. Blank lines separate every block (RTL-safe). No frontmatter.

Two frozen facts (must not be contradicted anywhere in this file):

Coupling best → worst = **Data > Stamp > Control > External > Common > Content** (Fig. 7, slide 19 / book p. 208–209).

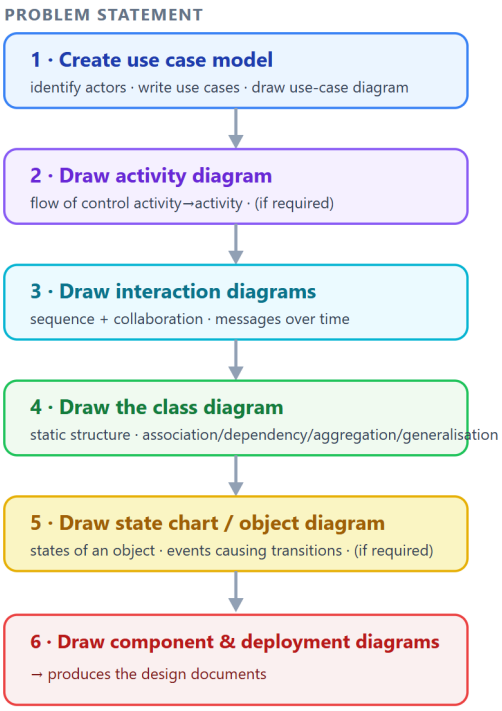
Cohesion best → worst = **Functional > Sequential > Communicational > Procedural > Temporal > Logical > Coincidental** (Fig. 11, slide 27 / book p. 211).

OOD = exactly **SIX** steps in this order (slide 72 Fig. 25 / book p. 232): (1) Create use case model → (2) Draw activity diagram (if required) → (3) Draw the interaction diagrams → (4) Draw the class diagram → (5) Draw the state chart / object diagram (if required) → (6) Draw component & deployment diagram → design documents.

Reference diagrams (review before solving)

The Six Steps of Object-Oriented Design

Fig. 25 · identical on slide 72 and in the book, printed p.232 · six steps, not seven



Why these six and no more

- Each step feeds the next: use cases define the actors, activity/interaction capture behaviour, the class diagram captures structure, state/component capture lifecycle & build.
- Steps 2 and 5 are marked "if required" — they are optional.

⚠ The exam trap

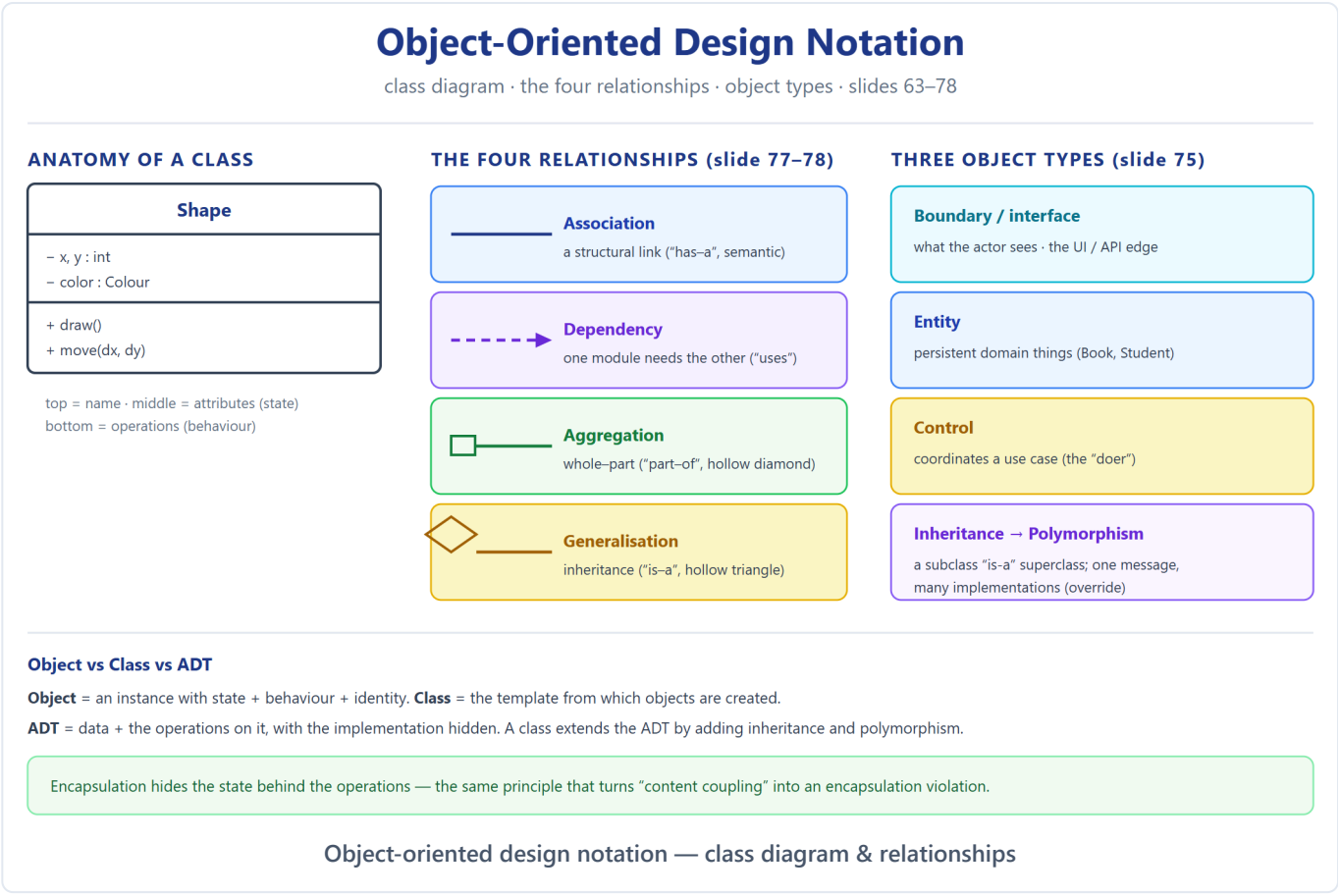
The slide (75) prints the interaction-diagram sub-steps as "a) b) d)" — it skips the letter (c). The chapter has no (c); the content is still exactly six stages. Do not invent a seventh.

Verified by re-rendering the book pages.

Diagram types produced (slide 75)

- Use-case diagram → actors & system boundary
- Activity diagram → dynamic control flow
- Sequence / collaboration → message exchange
- Class diagram → classes, attributes, operations
- State chart → object lifecycle
- Component / deployment → physical layout

The six steps of object-oriented design (Fig. 25)



Object-oriented design notation — class diagram & relationships

---

القسم أ — بناء نماذج الكائنات (Object / Class Artefacts)

Q6.1 (Create — class model with one relationship of each kind)

> Design the class model for an **Online Shop** system. It must contain **at least three classes** and **one relationship of each Chapter-5 kind**: association, dependency, aggregation, generalisation. Mark clearly the **polymorphic operation**.

Assumptions: single-language-agnostic model (slide 59 says OOD does not depend on an implementation language); a Customer places Orders; each Order contains OrderLines; payment is handled by an external PaymentGateway.

Artefact — class model (text form):

```
CLASS Customer
  Attributes : custId, name, email, address
  Operations : register(), login(), placeOrder()

CLASS Order
  Attributes : orderId, orderDate, status, total
  Operations : addLine(), removeLine(), checkout(), computeTotal()    <-- POLYMORPHIC
     subclass DiscountedOrder overrides computeTotal() (applies a coupon)
        subclass InternationalOrder overrides computeTotal() (adds duty)

CLASS OrderLine
  Attributes : productId, qty, unitPrice
  Operations : lineTotal()

CLASS Product
  Attributes : productId, name, price, stockQty
  Operations : isAvailable(), reserve(), release()

CLASS PaymentGateway    (external boundary)
   Operations : authorise(amount, token), refund(txnId)
```

:(78–per slides 77 ,one of each) relationships used — Artefact

Relationship	Chapter-5 definition (slide 77–78)	How it is used here	Notation
Association	"semantic connection between classes… each class can send messages to the other"	Customer ↔ Order (a customer places orders)	plain line
Dependency	"always unidirectional… one class depends on definitions in another"	Order ➜ PaymentGateway (checkout uses the gateway; gateway does not know Order)	dashed arrow
Aggregation	"a relationship between a whole and its parts"	Order ◆— OrderLine (whole = Order, parts = its lines)	diamond
Generalisation	"inheritance relationship between two classes"	DiscountedOrder ▷ Order (subclass inherits state/behaviour)	hollow triangle

Polymorphic operation mark: Order.computeTotal() — per slide 69, "we abstract just the interface of an operation and leave the implementation to subclasses" → each subclass supplies its own body, so the same message order.computeTotal() yields different behaviour by actual type.

التبرير العربي: المطلوب أربع علاقات — والفرق الحاسم بينها: الـ**association** رابط دلالي عادي، والـ**dependency** أحادي الاتجاه (يعتمد على تعريف الطرف الثاني)، والـ**aggregation** علاقة *كل/جزء* (الطلب يحتوي بنودًا)، والـ**generalisation** وراثة (subclass ترث من superclass). الـoperation polymorphic هي computeTotal() لأن واجهتها في الأب والتنفيذ في الأبناء (سلايد 69).

Q6.2 (Create — class model for an ATM, cohesion-aware)

> Design a class model for a **Bank ATM** with **at least four classes**. Explicitly separate **entity / boundary (interface) / control** objects (slide 75), and give exactly one **generalisation** and one **aggregation**.

Assumptions: a single bank; two account kinds; the ATM talks to a BankHost for debits.

Artefact:

```
== ENTITY classes (data that outlives the transaction) ==
CLASS Account            : acctNo, holderName, balance, type
                Operations: debit(), credit(), getBalance()
CLASS SavingsAccount   : minBalance, interestRate      (GENERALISATION ▷ Account)
CLASS CurrentAccount   : overdraftLimit                (GENERALISATION ▷ Account)
CLASS Card               : cardNo, pin, expiry, status
                Operations: validatePin(), isActive()
CLASS CashBin            : binId, denomMap                  (AGGREGATION owner)

== BOUNDARY (interface) classes ==
CLASS KeypadScreen     : Operations: readInput(), showMessage()
CLASS CardReader       : Operations: readCard()
CLASS CashDispenser    : Operations: dispense(amount)

== CONTROL classes (coordinate the use case) ==
CLASS TransactionController : Operations: withdraw(), deposit(), transfer()
```

Relationship map (wording 78–each must match slide 77):

Pair	Kind	Slide-definition check
SavingsAccount → Account	Generalisation	"inheritance relationship" (slide 78)
CashBin ◆— CashNote	Aggregation	"whole and its parts" (slide 78)
TransactionController ➔ Account	Dependency	"always unidirectional" (slide 78)
Card ↔ Account	Association	"semantic connection… each class can send messages" (slide 77)

Polymorphic operation: Account.debit() is overridden in SavingsAccount (enforces min balance) and CurrentAccount (allows overdraft).

التبرير العربي: تصنيف الكائنات الثلاثة إلزامي (سلايد 75): **entity** = بيانات دائمة (Account/Card)، **boundary** = واجهة مع المستخدم (Keypad/CardReader)، **control** = منسّق الـuse case (TransactionController). الفصل هذا هو تطبيق الـhiding information عمليًا: الـcontrol ما يوصل داخليات الـentity إلا عبر عملياتها.

القسم ب — عملية التصميم الكائني (The SIX OOD Steps)

Q6.3 (Create — realise the 6 OOD steps concretely)

> For the small problem "**an online cinema ticket booking system**", produce the chapter's **OOD steps as a concrete sequence**, naming the exact diagram produced at each step and the concrete content of that diagram.

Assumptions: one venue, one film per screen, seats selectable.

Artefact — the SIX steps (order and number are fixed by slide 72 Fig. 25):

#	Step (slide 72)	Diagram produced	Concrete content for this problem
1	Create use case model	**Use case diagram**	Actors: Customer, Payment Gateway. Use cases: Browse Films, Select Show, Select Seats, Pay, Print Ticket
2	Draw activity diagram (if required)	**Activity diagram**	Flow: browse → choose show → choose seats → [seats available?] → pay → issue ticket; the fork/branch models the availability check (slide 73: "an activity represents an operation on some class… that results in a change in the state")
3	Draw the interaction diagrams	**Sequence diagrams** (one per use case) + collaboration diagrams	e.g. Select Seats: Customer → SeatSelection : select(showId, seats); SeatSelection → SeatDB : checkAvailable(); SeatDB → Seat : markHeld()
4	Draw the class diagram	**Class diagram**	Classes: Film, Show, Seat, Booking, Customer; relationships per slides 77–78; Seat.price() polymorphic over StandardSeat/PremiumSeat
5	Draw the state chart / object diagram (if required)	**State chart diagram** + object diagram	State chart for Seat: Available → Held → Booked → (cancelled) Available (slide 79: "state space of a given class, the event that cause a transition")
6	Draw component & deployment diagram	**Component + deployment diagrams** → design documents	Components: BookingService, PaymentAdapter, SeatDB; deployment: DB server + app server + client browsers (slide 80)

التبرير العربي: الخطوات **ست بالضبط** بهذا الترتيب (سلايد 72) — كل خطوة تقيّد اللي بعدها: الـcase use يحدد الحدود، والنشاط يحدد التدفق، والتفاعل يحدد الرسائل (= الاقتران المقصود قبل البرمجة)، والصنف يثبّت الحالة والسلوك، ثم الحالة والنشر. خلط الترتيب يعني تصميم بنية قبل معرفة نطاقها.

Q6.4 (Create — contrast the 6 steps with a 5-step wrong answer)

> A student submits a 5-step OOD process: "(1) class diagram (2) use case model (3) sequence diagram (4) component diagram (5) write code." Grade it against the chapter and produce the corrected SIX-step version.

Assumptions: the question tests recall of slide 72 / book p. 232.

Artefact — grading table:

Student step	In chapter's 6-step list?	Defect
1. class diagram	Yes — but **position 4**, not 1	Classes are fixed *after* use cases and interactions; a class diagram drawn first models the wrong scope
2. use case model	Yes — **position 1**	Out of order; must be first (slide 73: "First step is to identify the actors")
3. sequence diagram	Yes — **step 3** (interaction diagrams)	Naming too narrow: slide 75 says interactions = sequence **and** collaboration diagrams
4. component diagram	Yes — **step 6**, and must be paired with **deployment**	Missing the deployment diagram
5. write code	**Not an OOD step**	Chapter step 6 ends at " → design documents"; coding is outside design

Corrected SIX-step sequence: (1) Use case model → (2) Activity diagram (if required) → (3) Interaction diagrams → (4) Class diagram → (5) State chart / object diagram (if required) → (6) Component & deployment diagram → design documents.

التبرير العربي: الطالب خلط الترتيب وحذف خطوة. القاعدة: الـ**use case أولاً** (يرسم الحدود)، والـ**class diagram رابعًا** (لأنه يحتاج نتائج التفاعل)، والخطوة السادسة **مزدوجة** (component **و** deployment). عدد الخطوات **ست** — مو خمس ولا سبع (سلايد 72).

القسم ج — وصف حالة الاستعمال (Use-Case Description)

Q6.5 (Create — full use-case description, mirroring "Issue Book")

> Write a **use-case description** for **"Issue Book"** in the Library Management System, mirroring the book's format: Introduction / Actors / Pre-condition / Post-condition / Basic flow / Alternate flow / Special requirements / Related use cases. Ground it in the requirements on slides 81–82 and the sequence diagram on slide 86.

Assumptions: standalone system (slide 81); bar-code detector available (slide 82); issue limits = max 4 Book-bank + 3 General for 15 days (slide 81).

Artefact — use-case description:

```
USE CASE : Issue Book                    (Library Management System)
------------------------------------------------------------------------
1. INTRODUCTION
  Allows an Operator to issue a book to a student using the bar-code
  detector. The system computes the return date (issue date + 15 days)
    and stamps the due date.

2. ACTORS
  Primary   : Operator
  Secondary : Bar code reader   (machine actor, slide 84 use case diagram)

3. PRE-CONDITION
  - Operator is logged in (Login use case completed).
    - Student detail and Book detail exist (Maintain student detail /
   Maintain catalogue completed).
    - Bar-code detector is online.

4. POST-CONDITION (success)
    - An Issued Book Details record is created (Book Id, Barcode Id, Issued).
    - A Student Issued Book Details record links the student to the book.
  - Book's "Copies in General / Book Bank" count is decremented.

5. BASIC FLOW
  1. Operator reads the student barcode.                (seq. 1)
    2. System submits student details to Student Info controller. (seq. 2)
    3. System fetches and validates student details.    (seq. 3-4)
    4. System checks the book can be issued to that student's course.
  5. Operator reads the book barcode.                   (seq. 8)
    6. System submits book code and re-checks issue rules. (seq. 9-10)
    7. System updates details, adds Issued Book record, adds Student Issued
    Book record, stamps the due date.                   (seq. 12-14)
    8. System reports "Issue successfully".             (seq. 15)

6. ALTERNATE FLOWS
  6a. Invalid student detail  -> show "Invalid student" (seq. 5); abort.
  6b. Book cannot be issued   -> show "Can't issue" (seq. 7); abort.
  6c. Book-bank book requested for a different course -> route to 6b.
  6d. Issue limit reached (4 Book-bank / 3 General)   -> show limit error.

7. SPECIAL REQUIREMENTS
    - Issue period is fixed at 15 days; due date = system date + 15.
    - Due date is physically stamped on the book (slide 82).
    - Bar-code input is mandatory for both student and book.

8. RELATED USE CASES
    <<include>> Login ;  <<extend>> Print receipt ;
    <<uses>> Maintain student detail, Maintain catalogue.
```

التبرير العربي: الوصف يطابق بنية الكتاب المطلوبة، وكل سطر مُشتق من سلايد 81–82 + تسلسل سلايد 86. الفائدة الهندسية: هذا المستند هو الحدّ الفاصل بين النطاق والداخل — الـflows alternate تمنع تكديس الشروط داخل الـflow basic (نفس فكرة تجنّب logical cohesion على مستوى النموذج).

Q6.6 (Create — write the Return-Book use case, then justify its separation)

> Write the use-case description for **"Return Book"** (slide 82 requirements, slide 87 sequence) and then justify — using cohesion principles — why the book keeps it **separate** from Issue Book.

Assumptions: any person may return the book (slide 82); a fine may apply.

Artefact — concise use-case description:

Section	Content
Introduction	Accepts a returned book and updates the database.
Actors	Primary: Operator; Secondary: Bar code reader
Pre-condition	Book was issued; operator logged in; detector online
Post-condition	Issued Book record deleted; copies incremented; fine charged if late
Basic flow	1 read code → 2 submit → 3 validate if fine → 4 calculate fine → 5 update Student details → 6 delete Student-Issued-Book record → 7 update Issued-Book-Details → 8 "Book returned successfully"
Alternate flow	8a. No fine due → skip step 4, proceed to 5. 8b. Unknown book code → show error, abort.
Special requirements	Fine computed from issue date vs return date (15-day rule)
Related use cases	<<include>> Login ; <<uses>> Query book

Why separate from "Issue Book" (cohesion argument): the two flows have **different pre/post-conditions and different actors** — one creates an issue record, the other deletes it and may charge a fine. Merging them into one "Circulation" artefact would create a **single description whose behaviour branches on a control condition (issue vs return)** — the *model-level equivalent of logical cohesion / control coupling*. Keeping them separate gives each description **a single, well-defined purpose** (the model equivalent of functional cohesion), which is easier to read, verify and implement.

التبرير العربي: الفصل لأن الوصفين مختلفون في الشروط والأحداث والأخطاء. دمجهما يعطينا وصف واحد يتفرّع على راية (issue/return) — نظير الـ**logical cohesion** الوارد في سلايد 30. الإبقاء منفصلين يعطي كل use case **غرضًا واحدًا واضحًا** = نظير الـfunctional cohesion.

القسم د — الوحدات والاقتران والتلاحم (Modular Decomposition)

Q6.7 (Create — decompose a system, classify every cohesion and coupling)

> Take the requirement set: "**The system must register a student, record a course, enrol a student in a course, and produce a transcript**". Decompose it into modules and, for **each module**, state and justify its **cohesion type** and its **coupling type** to its callers (target: high cohesion, low coupling, slide 31).

Assumptions: persistent store accessed only through parameterised calls; no global record buffer.

Artefact — module decomposition with classification:

```
MAIN  (transaction dispatcher)
  ├── register_student(studentRecord)  :  void
  ├── record_course(courseRecord)      :  void
  ├── enrol_student(studentId, courseId)  :  Result
  ├── generate_transcript(studentId)   :  Transcript
  └── storage_layer   (accessed only via parameters)
```

Module	Cohesion	Justification	Coupling (to callers)	Justification
register_student	**Functional**	single task: create + store one student record	**Data**	receives one primitive-structure parameter, returns status
record_course	**Functional**	single task: create + store one course record	**Data**	one parameter
enrol_student	**Functional**	single task: create the enrolment link after validating both keys	**Data**	two ids in, result out — no flag (flag would make it **Control**)
MAIN dispatcher	**Logical** (weak)	it selects *which* task to run on the same logical class "handle request"	**Data**	passes only the request id; branching stays inside MAIN, not signalled by a flag from above
generate_transcript	**Sequential**	fetch enrolments → fetch grades → format; each step feeds the next (slide 28)	**Data**	studentId in; transcript out
storage_layer	**Communicational**	all operations work on the same persistent store (same input/output resource)	**Data**	every access is a parameterised call — explicitly **no global buffer**, so **no Common coupling** (slide 21)

Design verdict: **Communicational** → **Sequential** → **Functional** cohesion ranges (good end of Fig. 11); sits at Logical and MAIN there is no shared global area. Only **Data** coupling is throughout (best end of Fig. 7). grows should be split further if it.

التبرير العربي: الهدف على سلايد 31: **low coupling + high cohesion**. هنا كل وحدة وظيفية (Functional) ما عدا الـdispatcher (Logical) والـtranscript (Sequential لأنها سلسلة). الاقتران كله **Data** (الأفضل) لأن المدخلات parameters — وتجنّبنا الـ**Common coupling** بعدم استخدام أي global buffer (سلايد 21).

Q6.8 (Create — design to a stated coupling budget)

> Design a **Payroll** module set with **no coupling worse than Control** and **no cohesion worse than Procedural**. List your modules, their cohesion, their coupling, and state the one decision that binds you to the budget.

Assumptions: monthly run; tax rules in a separate table; employee master file available.

Artefact:

Module	Cohesion	Coupling	Contract
compute_gross(empRecord)	Functional	Data	one employee record in; gross out
`apply_tax(gross, taxTable)	Functional	Data	passes the tax table **by parameter** (never reads a global)
run_payroll(masterFile)	Sequential	Data	drives compute → tax → slip in order
print_payslip(empRecord, gross, tax)	Procedural	Data	three data parameters, no flags
select_employees(filterMode)	**Logical**	**Control**	filterMode is a **control flag** → this is the one permitted Control coupling (slide 20)

Budget-binding decisions:

No globals — the tax table is passed as a parameter, keeping everything at Data (slide 21: "making a change to the common data means tracing back to all modules which access that data").

The only Control link is quarantined in select_employees, where the flag genuinely selects behaviour; the budget forbids it anywhere else.

Cohesion floor honoured: the weakest module is select_employees at Logical — if the budget is enforced strictly, split it into select_by_department / select_by_grade, which raises it to Functional.

التبرير العربي: الميزانية تسمح بأقصى coupling **Control** وأدنى cohesion **Procedural**. حاصرنا الـcontrol في وحدة واحدة (select_employees) لأن الراية هناك اختيار سلوك فعلي (سلايد 20). تجنّب الـglobals هو القرار الحاسم — أول ما تدخل global area يصير الاقتران **Common** ويُخرق السقف (سلايد 21).

Q6.9 (Create — refactor a bad decomposition to target levels)

> Given: "one utility_module contains 12 unrelated helpers, and every other module reads a global config_string." Produce the **refactored** decomposition and state the **before/after** level on both scales.

Assumptions: the 12 helpers cluster into 3 real functions; the config values are 5 primitives.

Artefact — refactor plan:

```
BEFORE                          AFTER
--------                                    -----
utility_module (12 helpers)        -->      file_io_utils          (Functional)
global config_string  <-- read              validation_utils       (Functional)
   by every module                          math_utils             (Functional)
                                            config = 5 params passed explicitly
```

Aspect	Before	After	Scale movement
utility_module cohesion	Coincidental (worst)	Functional (best)	Fig. 11: 7/7 → 1/7
Modules reading config_string	Common coupling (worst-but-one)	Data coupling (best)	Fig. 7: 5/6 → 1/6
Rule applied	—	split by function; pass 5 primitives	slide 31 objective

التبرير العربي: الخلل المزدوج: وحدة بلا صلة (Coincidental) + حالة عامة مشتركة (Common coupling). الحل المزدوج: تفكيك الوحدة حسب الوظيفة (Functional) + تمرير قيم الإعداد كـparameters (Data). هذا هو بالضبط معنى **high cohesion / low coupling** في سلايد 31، ويقلّل الـripple (سلايد 21).

القسم هـ — مخطط البنية (Structure Chart)

Q6.10 (Create — structure chart and its distinction from a flowchart)

> Draw/describe a **structure chart** for "**Update File**" (slide 43, Fig. 18) and state the **three differences** between a structure chart and a flowchart.

Assumptions: record-oriented update; validation before write.

Artefact — structure chart as an indented tree (slide 41: "partition a system into block boxes... black box = functionality known without internal design"):

```
UPDATE FILE   (root / level 0)
├── INPUT
│       reads record key
│       (data pass: recordKey  downward)
├── VALIDATE
│       checks key against file
│       (data pass: recordKey  →  validFlag)
├── UPDATE-RECORD
│       ├── READ-RECORD        (data: recordKey  →  record)
│       ├── MODIFY-FIELDS      (data: newValues  →  record)
│       └── WRITE-RECORD       (data: record  →  status)
└── OUTPUT
    prints status message          (data: status)
```

:structure chart vs flowchart — Artefact

Aspect	Structure chart (slide 41–42)	Flowchart
Shows	**Hierarchy of modules / who calls whom** (static)	**Sequence of processing steps** (dynamic order)
Unit (box)	A **module** — a black box with a known interface	A **process step / decision**
Annotation	Data and control **couples on the call arrows** (Fig. 17, slide 42)	Flow lines between steps

الفرق الجوهري: الـstructure chart يُظهر **التسلسل الهرمي للوحدات وماذا يُمرَّر بينها** (بيانات/تحكم) — فهو ثابت (static) ويخدم تحليل الاقتران. الـflowchart يُظهر **ترتيب خطوات المعالجة** داخل الوحدة (dynamic). الأول يجيب "من ينادي من"، الثاني يجيب "ما هو ترتيب التنفيذ".

Q6.11 (Create — transaction-centred structure chart)

> Design a **transaction-centred structure chart** (slide 44, Fig. 19) for a **library front desk** that processes several transaction types, and state the three responsibilities of the MAIN module (slide 45).

Assumptions: transaction types = Issue, Return, Query, Report.

Artefact — transaction-centred structure:

```
LIBRARY-FRONT-DESK (MAIN)
│   (1) invoke INPUT to read a transaction
│   (2) determine the kind of transaction, select one transaction module
│   (3) output results by calling OUTPUT
├── INPUT                    (reads the transaction record)
├── TRANSACTION-DISPATCH  (selects one of the below on transaction type)
│       ├── ISSUE-BOOK
│       ├── RETURN-BOOK
│       ├── QUERY-BOOK
│       └── GENERATE-REPORT
└── OUTPUT                   (prints the transaction result)
```

:(verbatim from slide 45) MAIN's three functions (i) invoke the INPUT module to read a transaction ;(ii) determine the kind of transaction and select one of a number of transaction modules to process it ;(iii) output the results by calling the OUTPUT module.

Coupling note: the dispatcher passes the **transaction record by data**; if instead it passed a typeFlag that the leaf modules branched on, coupling would rise to **Control** (slide 20). The chart above keeps the branch inside the dispatcher, so every leaf stays **Data**-coupled.

التبرير العربي: البنية **transaction-centred** (سلايد 44) = بنية تدور على محول يختار الوحدة حسب نوع المعاملة. مسؤوليات MAIN الثلاث من سلايد 45 (input، تحديد النوع والتوزيع، output). إبقاء التفرّع داخل المُوزِّع يحافظ على اقتران **Data** للأوراق ويمنع ارتفاعه إلى **Control**.

القسم و — مستند تصميم البرمجية (SDD)

Q6.12 (Create — SDD outline with concrete artefacts per view)

> Write the **SDD outline** for the Library Management System using IEEE STD 1016-1998: the **four design views** of Table 2 (slide 57) with the entity attributes each view uses, and choose a **concrete artefact** for each view. Then show how the ten entity attributes (a–j, slide 53) map onto one entity.

Assumptions: SDD organisation follows slide 52 (Introduction / Design entities / Design entity attributes).

Artefact — four design views (Table 2, slide 57):

Design view	Scope (slide 57)	Entity attributes used	Concrete artefact for Library System
Decomposition description	Partition of the system into design entities	Identification, type, purpose, function, subordinate	Hierarchical decomposition diagram + module list: Circulation → Issue Book, Return Book, Query Book
Dependency description	Relationships among entities of system resources	Identification, type, purpose, dependencies, resources	Structure chart showing how Issue Book calls Validate Student, Check Issue Rules, Update DB
Interface description	Everything needed to *use* each entity	Identification, function, interfaces	Parameter table for `issueBook(studentBarCode, bookBarCode) : Status`
Detail description	Internal design details of an entity	Identification, processing, data	PDL/pseudocode of the fine calculation in Return Book

Artefact — the ten entity attributes (a–j, slide 53) applied to the entity Issued Book Details:

Letter	Attribute (slide 53)	Value for Issued Book Details
a	Identification	E-04 Issued Book Details
b	Type	Entity (persistent data)
c	Purpose	Record which book is issued to a student
d	Function	Issue / Return / Add / Delete
e	Subordinates	none (leaf entity)
f	Dependencies	Book Details, Student Issued Book Details
g	Interface	`Add(), Delete(), Issue(), Return()` (slide 93)
h	Resources	Library database, bar-code reader
i	Processing	stamp due date = issue date + 15 days
j	Data	Book Id, Barcode Id, Issued

التبرير العربي: الـSDD = **blueprint للتنفيذ** (سلايد 52). أربع views (سلايد 57) وكل view يستخدم مجموعة صفات مختلفة؛ وعشر صفات للـentity من a إلى j (سلايد 53). القاعدة العملية: الـdecomposition يقول "ما هي الأجزاء"، والـdependency "كيف تتصل"، والـinterface "كيف تُستعمل"، والـdetail "كيف تعمل من الداخل".

القسم ز — مهام إنشاء إضافية (Additional Create Tasks)

Q6.13 (Create — interaction diagram description for a named use case)

> Produce a **sequence diagram description** for the Library "**Query Book**" use case, naming the object types (slide 75) and the messages (slide 88).

Assumptions: user may be student, librarian or operator (slide 88 note).

Artefact — sequence (text form), objects typed per slide 75:

```
:User                    (actor)
:Query book interface    (BOUNDARY / interface object)   – lifeline 1
:Query operator          (CONTROL object)                – lifeline 2
:Book details            (ENTITY object)                 – lifeline 3

1. User                -> :Query book interface : enter name/author
2. :Query book interface -> :Query operator : get details
3. :Query operator     -> :Book details     : search details
4. :Book details       -> :Query operator   : result
5. :Query operator     -> :Query book interface : show result  (to User)
```

Book ;control = Query operator ;interface/boundary = Query book interface **:(slide 75) Object-type check** one ,all show one boundary 92–slides 85) case study entity. This is the same pattern used across the = details .(control and one entity per use case

التبرير العربي: كل use case في حالة الدراسة له ثلاثة أنواع كائنات: **boundary** (النموذج/الواجهة)، **control** (المتحكّم)، **entity** (البيانات). نمط الرسائل موحّد: المستخدم → الحدّ → المتحكّم → الكيان ثم الرجوع. هذا يعطي اقترانًا أحاديّ الاتجاه واضحًا قبل كتابة الشيفرة (سلايد 75).

Q6.14 (Create — build an entity class diagram from the case study)

> From the requirements (slides 81–83), construct the **class diagram of entity classes** for the Library System and identify each relationship's kind per slides 77–78.

Assumptions: entity classes only (slide 93 shows the book's own answer).

Artefact — entity classes (matches slide 93):

```
CLASS Book Details
 Book Id, Book Name, Author Name, Publisher,
  Copies in Book Bank, Copies in General
 Add Book(), Update No. of Copies()

CLASS Student Details
 Student Barcode Id, Student Roll no., Name, Course,
  No. of Book Bank Issued, No. of General Issued
 Get Student Details(), Add Student(), Delete Student(), Update Student()

CLASS Issued Book Details
 Book Id, Barcode Id, Issued
 Add(), Delete(), Issue(), Return()

CLASS Student Issued Book Details
 Student Bar Code Id, Book Bar Code Id
 Add(), Delete()

CLASS Login Details
 User Id, User Name, Password, User Role
 Add(), Delete(), Update()
```

:relationships — Artefact

From	To	Kind	Slide-78/77 wording check
Issued Book Details	Book Details	Dependency (unidirectional)	"always unidirectional… depends on definitions in another"
Student Issued Book Details	Student Details	Dependency (unidirectional)	same
Issued Book Details ↔ Student Issued Book Details	Association	"semantic connection between classes" (slide 77)	

التبرير العربي: المخطط طابق ما في سلايد 93 حرفيًا. العلاقة بين تفاصيل الكتاب الصادر وتفاصيل الكتاب = **dependency** (أحادية الاتجاه، سلايد 78)، والربط بين الكتاب الصادر والطالب = **association** (سلايد 77). هذا يفصل "بيانات الكيان" عن "سجل الحركة" بفصل نظيف.

Q6.15 (Create — justify an OOD choice with a trade-off)

> Produce a **design-decision memo** (max one page) justifying **object-oriented design over function-oriented design** for the Library System, using at least four chapter concepts, and state the single risk of your choice.

Assumptions: requirements (issue limits, fine rules) are expected to change after deployment.

Artefact — memo:

```
DESIGN DECISION MEMO — Library Management System
DECISION: Adopt object-oriented design (6-step process, slide 72).

1. ORTHOGONALITY (slide 58): OOD focuses on the data ("things"), not the
   function; the library's things (Book, Student, Loan) are stable while
   the functions change -- so the decomposition is more durable.

2. ENCAPSULATION / INFORMATION HIDING (slide 69): each entity keeps its own
   state; the fine rule lives inside the Loan class, so changing it does
   not ripple to unrelated modules.

3. INHERITANCE + POLYMORPHISM (slides 68-69): Book Bank Book and General
   Book are subclasses of Book; the issue-limit rule becomes a polymorphic
   operation, so a new book category is added by subclassing -- no edit to
   existing code.

4. USE-CASE-DRIVEN TRACEABILITY (slides 73-75): the 6 steps start from the
   use case model, so every class traces to a requirement -- protecting the
   design from scope creep (the model-level source of coincidental cohesion).

RISK: OOD adds modelling overhead and requires discipline; if the team
documents classes but not their interfaces, class coupling can silently
become Content coupling -- the exact defect the chapter ranks worst.
```

التبرير العربي: اخترنا OOD لأن **الأشياء ثابتة والوظائف متغيّرة** (سلايد 58)، والـ**encapsulation** تحصر أثر التغيير (سلايد 69)، والـ**inheritance/polymorphism** تضيف سلوكًا بدون تعديل الكود القائم. الخطر الوحيد: إهمال توثيق الواجهات يخلق **Content coupling** — أسوأ نوع (سلايد 24).

Q6.16 (Create — assemble a complete mini-design for a new system)

> For "**a university hostel mess billing system**", produce a **complete mini-design**: the 6 OOD steps, one class model with ≥3 classes, one use-case description, and the cohesion/coupling of the modules you propose.

Assumptions: monthly billing; students have meal plans; cashier records payments.

Artefact — (A) the SIX OOD steps:

#	Step	Diagram	Content
1	Use case model	Use case diagram	Actors: Cashier, Student. Use cases: Enrol Student, Record Meal, Generate Bill, Record Payment, Query Dues
2	Activity diagram (if req.)	Activity diagram	Flow: select student → check plan → count meals → compute bill → [paid?] → close / carry forward
3	Interaction diagrams	Sequence diagrams	Generate Bill: Cashier → BillForm → BillController → MessRecordDB
4	Class diagram	Class diagram	Student, MealPlan, Bill, Payment

#	Step	Diagram	Content
5	State chart / object diagram (if req.)	State chart	Bill: Open → Generated → PartlyPaid → Settled
6	Component & deployment	Component + deployment	Components: BillingService, PaymentAdapter, MessDB; deployment: campus server + cashier terminals

Artefact — (B) class model:

```
CLASS Student          : rollNo, name, hostelBlock ; enrol(), getPlan()
CLASS MealPlan         : planId, mealsPerDay, rate    ; monthlyAmount()    <-- POLYMORPHIC
    subclass VegPlan    overrides monthlyAmount()
        subclass NonVegPlan  overrides monthlyAmount()
CLASS Bill             : billId, month, amount, status ; generate(), settle()
CLASS Payment          : payId, amount, mode, date     ; record()
```

MealPlan ➜ Bill ;(whole/parts of a student's plan — **aggregation**) MealPlan —◆ Student :Relationships ▷ NonVegPlan/VegPlan ;(**association**) Bill ↔ Student ;(needs the plan to compute the amount — **dependency**) .(**generalisation**) MealPlan

Artefact — (C) use-case description (abridged, "Record Payment"):

```
Introduction    : Records a student's mess payment against an open  bill.
Actors          : Primary Cashier ; Secondary none
Pre-condition   : Bill exists with status Generated or PartlyPaid
Post-condition  : Payment recorded; bill status advanced or Settled
Basic flow      : 1  enter rollNo  -> 2  system shows open bill  -> 3  enter amount
          and mode  -> 4  system validates  -> 5  system records payment
                  -> 6  shows receipt
Alternate flow  : 4a  amount exceeds dues  -> error, return to 3
          4b no open bill  -> show message, abort
Special req.    : partial payments allowed; receipt is printed
Related         : <<include>> Login ; <<uses>> Query Dues
```

:module classification (D) — Artefact

Module	Cohesion	Coupling
count_meals(studentId, month)	Functional	Data
compute_bill(studentId, month)	Sequential	Data
record_payment(billId, amount, mode)	Functional	Data
generate_bill_report(month)	Communicational	Data

التبرير العربي: التصميم يمر بالخطوات **الست** بالترتيب، والـmodel class فيه علاقة من كل نوع (aggregation = plan/student، dependency = bill/plan، association = student/bill، generalisation = veg/nonveg plan)، والوحدات كلها **Functional/Sequential** بتلاحم عالٍ واقتران **Data** منخفض — أي التزام بهدف سلايد 31.

ملخّص التحقق (Verification Summary)

Check	Requirement	Status in this file
1	OOD steps = exactly **6**, correct order	Q6.3, Q6.4, Q6.16 all list **six** in the slide-72 order (use case → activity → interaction → class → state → component/deployment)
2	Relationship names match slide definitions	Q6.1, Q6.2, Q6.14 quote slide 77–78 wording: aggregation = whole/parts; generalisation = inheritance; dependency = unidirectional; association = semantic connection
3	Cohesion/coupling classifications match verified definitions	All classifications in Q6.7–Q6.9, Q6.16 use the Fig. 7 and Fig. 11 orders and the slide-28/29/30 / slide-20/21 definitions
4	Source citations	Every item cites slide (and book page where relevant)

State uncertainties (explicit):

Slides 84–93 are **image-only** (no extractable text); their content above was read by rendering the pages at 140 dpi and reading the figures directly. The use-case names and sequence messages are therefore transcribed from images and could contain a minor OCR-style variant in wording (e.g. "Get Student Details()").

Slide 75 lists interaction-diagram steps as "a) b) d)" — the letter **c) is skipped in the slide text** but the content is complete (the source map confirms the book uses a→f). This file uses the six *boxes* from Fig. 25 (slide 72), which are unambiguous.

The book's printed pp. 236–247 for the case study were **not opened directly** (only the slide deck was read); all case-study facts here are grounded in slides 81–93.

المصدر: Singh & Aggarwal, Software Engineering 3rd ed., Ch.5 — سلايدات 1–99 (التركيز 72–93) + كتاب ص.202–251.
كل صنف/علاقة/خطوة مُتحقَّق منه بنص الفصل.